

Introduction to Macroeconomics (S102016CR)

Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

Marcelo Olarreaga, Office M5270

```
graph TD; A[Introduction to Macroeconomics] --> B[Part I  
National Accounts and the Real Economy  
(Weeks 1 to 3)]; A --> C[Part II  
Monetary Economics and Financial Economics  
(Weeks 4 to 6)]; A --> D[Part III  
Open Macroeconomics  
(Weeks 7 to 9)]; A --> E[Part IV  
Macroeconomic Policies  
(Weeks 10 to 12)];
```

Introduction to
Macroeconomics

Part I
National Accounts and
the Real Economy
(Weeks 1 to 3)

Part II
Monetary Economics and
Financial Economics
(Weeks 4 to 6)

Part III
Open Macroeconomics
(Weeks 7 to 9)

**Part IV
Macroeconomic Policies
(Weeks 10 to 12)**

Fourth Part:
Macroeconomic
Policies

```
graph TD; A[Fourth Part: Macroeconomic Policies] --> B[AD and AS model (Chapter 10)]; A --> C[Government Policies (Chapter 11)]; A --> D[Inflation and Unemployment (Chapter 12)];
```

**AD and AS
model
(Chapter 10)**

Government
Policies
(Chapter 11)

Inflation and
Unemployment
(Chapter 12)

Week 10:
Short-run economic
fluctuations: AD and AS

In the long run, the capacity to produce is given by factor endowments and technology.

The classical dichotomy teaches that monetary policy will have no impact on real output in the long run. Its only impact will be on prices and therefore nominal output. Fiscal policy will not have an impact on real output in the long run either.

But in the short-run, economies are not always at their long-run equilibrium, as frictions prevent the full adjustment to long-run equilibrium after a shock.

In this chapter, we present the aggregate demand and aggregate supply model of the short-run. The model confirms the Keynesian idea that government spending is an essential tool to help economies reach their long-run full employment equilibrium when affected by negative shocks.

Monetary policy can also help, as we will see in the next two chapters

Chapter Contents

- a) Short-run fluctuations
- b) Aggregate Demand
- c) Aggregate supply
- d) Two main causes of short-run fluctuations
- e) Summary

Chapter Contents

a) Short-run fluctuations

b) Aggregate Demand

c) Aggregate supply

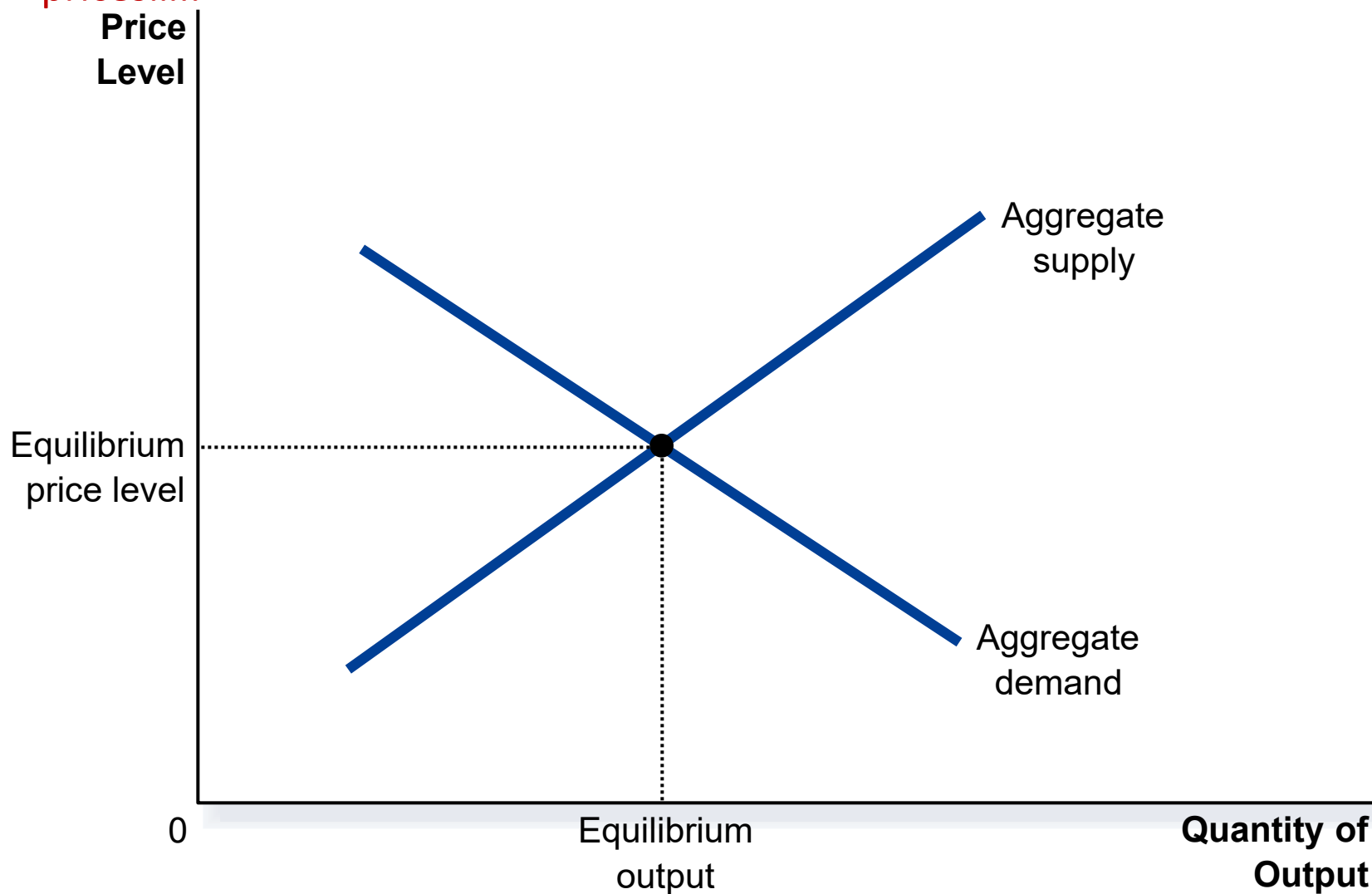
d) Two main causes of short-run fluctuations

e) Summary

a. Short-run fluctuations

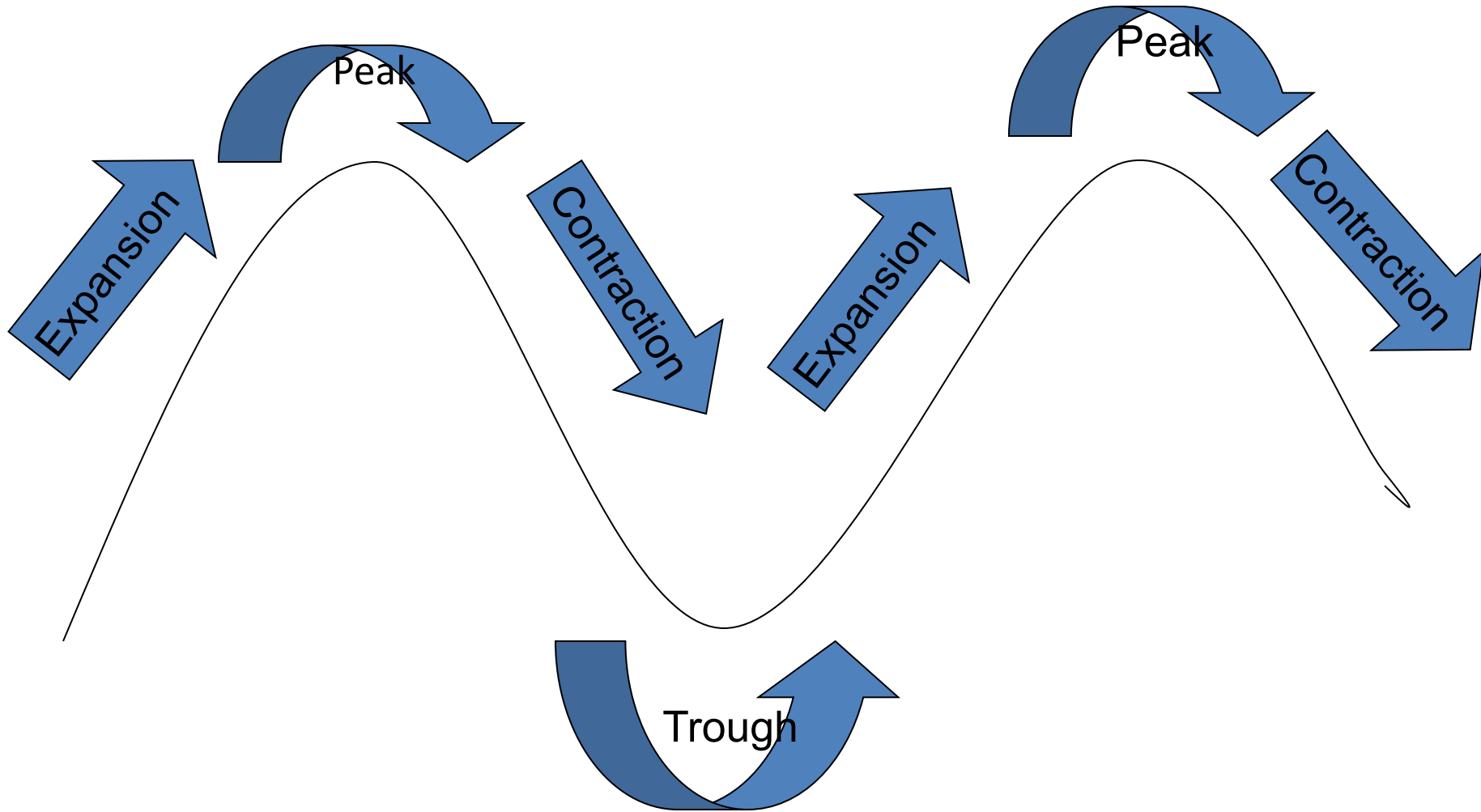
- Economic activity fluctuates from year to year.
 - In most years, the production of goods and services rises.
 - In some years, normal growth does not occur, causing a reduction in economic activity (sometimes a recession or even a depression).
- A *recession* is a period of declining real incomes and rising unemployment.
- A *depression* is a severe recession (i.e., a recession lasting more than two years or leading to GDP declining above 10 percent).
- Economic fluctuations are irregular and unpredictable.
 - Fluctuations in the economy are often called the business cycle.
- Most macroeconomic variables fluctuate together: as output falls, unemployment rises.
- To study these short-run fluctuations, we will develop a model of aggregate demand and supply that determines two variables: the real output of the economy measured by real GDP and the overall price level measured by CPI or the GDP deflator.

Aggregate Demand and Aggregate supply determine real GDP and overall prices.....



But what determines aggregate demand and supply?

Shifts in AD and AS explain the business cycle



Chapter Contents

a) Short-run fluctuations

b) Aggregate Demand

c) Aggregate supply

d) Two main causes of short-run fluctuations

e) Summary

b. Aggregate demand

- The four components of GDP (Y) contribute to the aggregate demand for goods and services.

$$Y = C + I + G + NX$$

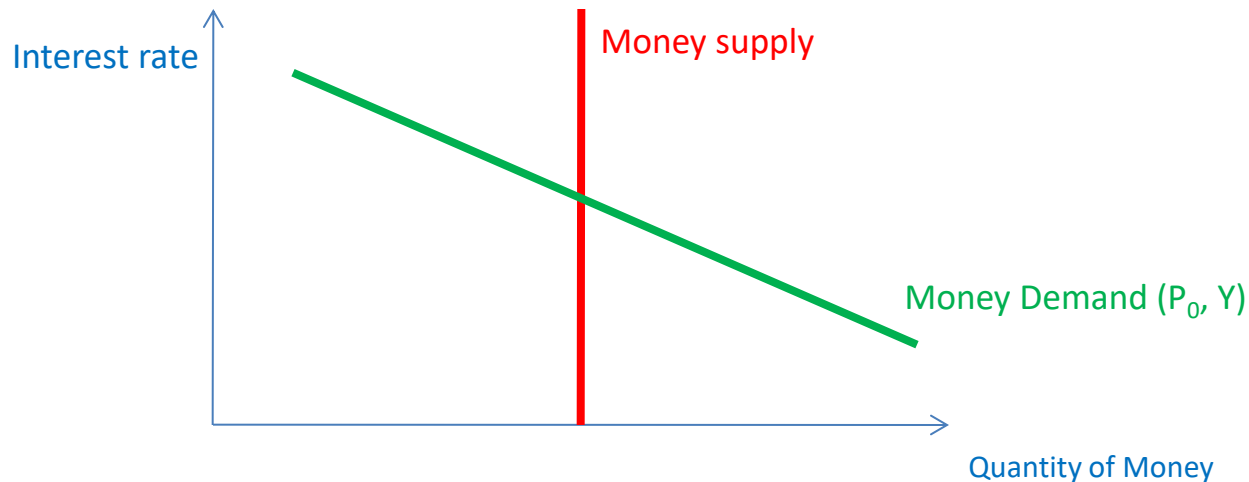
- Aggregate Demand will decline with overall prices, as seen in the micro class, but for different reasons. In microeconomics, demand increases when prices fall because of the “substitution” and “income” effects (putting aside Giffen goods).
- At the macro level, there is no room for “substitution” effects. As we are talking about “aggregate” demand. Nothing to substitute.
- No room for “income” effects as real income is what we determine
- At the macro level, aggregate demand falls with higher prices for three reasons:
 - The Price Level and Consumption: The Wealth Effect
 - The Price Level and Investment: The Interest Rate Effect
 - The Price Level and Net Exports: The Exchange Rate Effect

The Wealth Effect

- A decrease in the price level makes consumers feel wealthier (can buy more goods & services with the money they have at the bank), which in turn encourages them to spend more.
- This increase in consumer spending means larger quantities of goods and services demanded.

The Interest Rate Effect

- A lower price level reduces the interest rate because of the theory of liquidity preferences

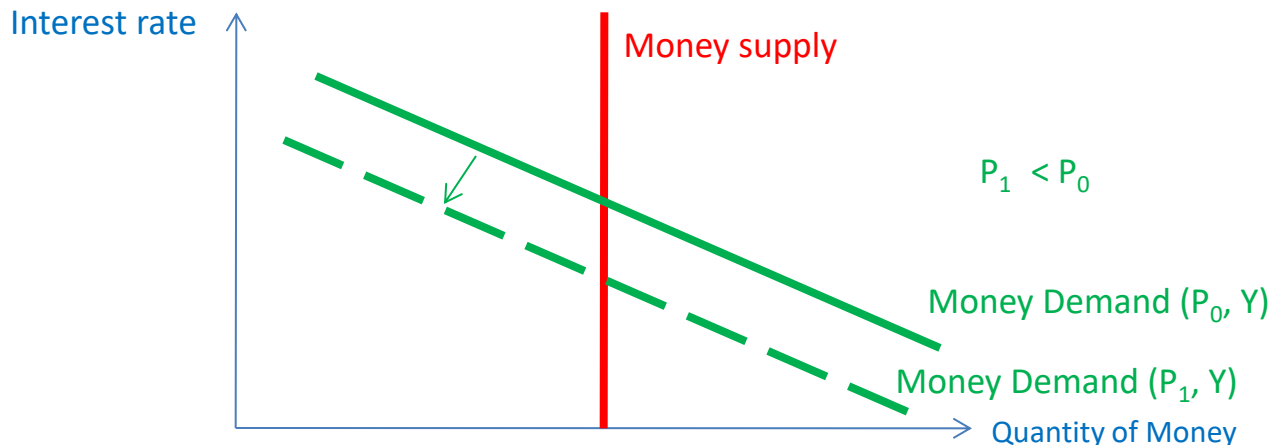


The Wealth Effect

- A decrease in the price level makes consumers feel wealthier, which in turn encourages them to spend more.
- This increase in consumer spending means larger quantities of goods and services demanded.

The Interest Rate Effect

- A lower price level reduces the interest rate, because of the [theory of liquidity preferences](#)



- This encourages investment and consumption and therefore aggregate demand

The Exchange Rate Effect

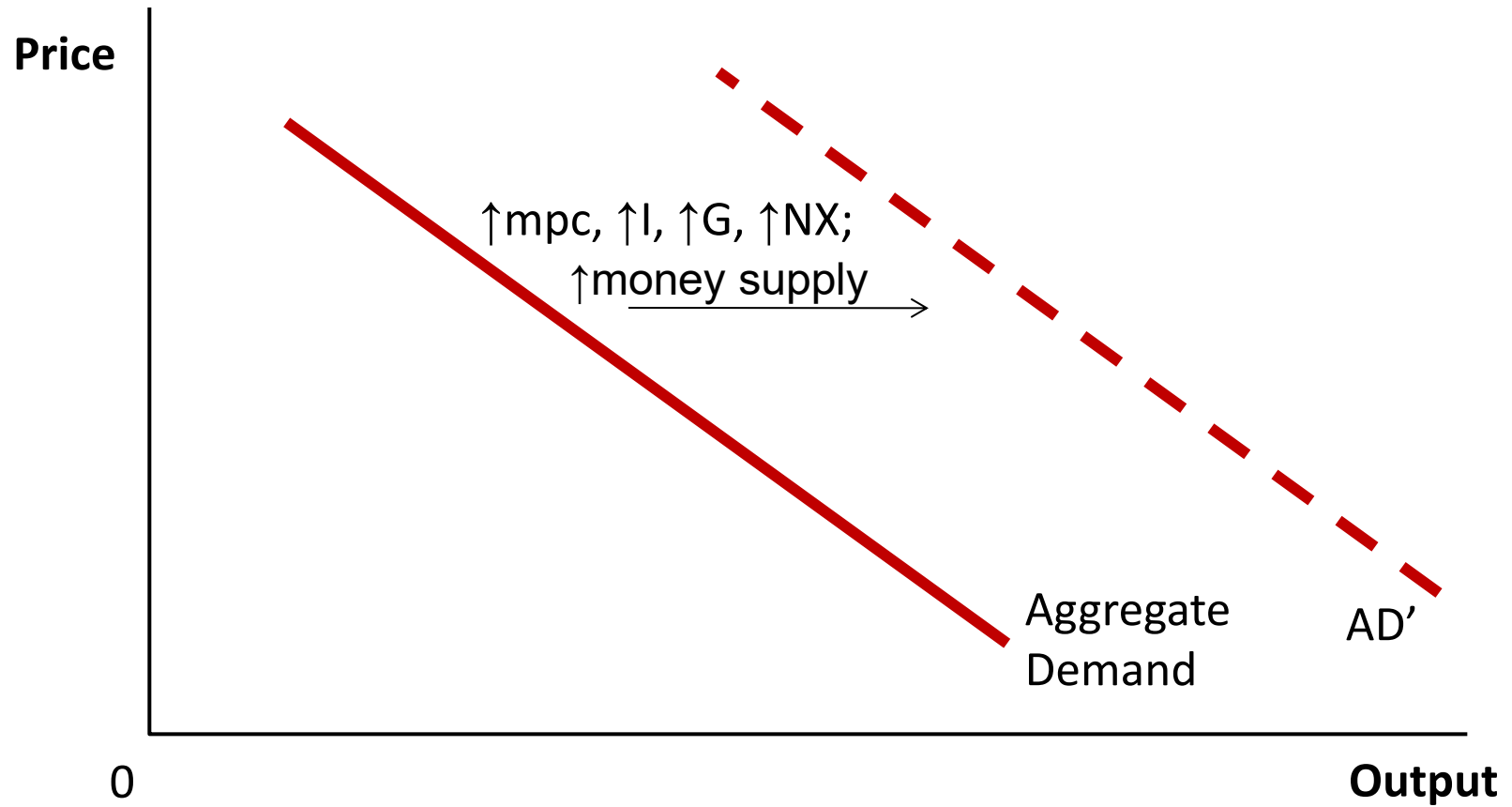
- A fall in home's overall prices leads to a depreciation of the domestic currency and thereby an increase in net exports: Why?
- Because as the overall price level falls, this leads to a decline in interest rates (theory of liquidity preferences), which leads to an increase in net capital outflows, which increases the demand for foreign currency, and leads to a depreciation of the domestic currency in real terms, i.e., an increase in home's real exchange rate (see Week 10).
- This depreciation of the real exchange rate makes home goods cheaper relative to foreign goods and will lead to an increase in net export spending means a larger quantity of goods and services demanded by the rest of the world.
- You can think of the exchange rate effect as a sort of substitution effect with goods produced in the rest of the world...
- For a very open economy, such as CH, the most important reason for the downward slope of the aggregate demand curve is the exchange rate effect. The interest rate effect is the most important reason for a relatively closed economy like the US.

So aggregate demand will slope downwards.

But we will also observe **shifts** in aggregate demand due to:

1. Changes in consumers' willingness to consume rather than save (more consumption-oriented individuals). An increase in life expectancy, expectations of a recession or a boom, or a stock market bubble or bust could cause this. Expectations are what Keynes called « animal spirits ». To capture their importance in a few minutes, there is no better than Olivier Blanchard's short video: http://voxeu.org/content/self-fulfilling-pessimism#.WLV_BFCECyA.twitter
2. Changes in the willingness to invest because of shocks to expectations regarding future growth
3. Changes in the size of the government
4. Changes in the rest of the world's demand for our goods due to changes in taste or business cycles in the rest of the world

Shifts in AD



Chapter Contents

- a) Short-run fluctuations
- b) Aggregate Demand
- c) Aggregate supply
- d) Two main causes of short-run fluctuations
- e) Summary

c. Aggregate supply

- In the **long run**, the *aggregate supply* curve is *vertical*.
 - The price level does not affect the aggregate supply in the long run.
 - In the long run, an economy's production of goods and services depends on its supplies of labor, capital, and natural resources and on the available technology used to turn these factors of production into goods and services.
 - These variables do not depend on aggregate prices
 - The long-run aggregate supply curve is vertical at what is called the natural rate of output.
 - This production level is also called potential output or full-employment output.
- Any change in the economy that alters the natural rate of output shifts the long-run aggregate supply curve.
- The shifts may be categorized according to the various factors in the classical model that affect output (labor, human and physical capital, natural resources, or technology).

- In the **short run**, the **aggregate** supply curve is *upward-sloping*.
 - A decrease in the level of prices tends to reduce the quantity of goods and services supplied.
 - Three theories to explain this:
 - The Sticky Wage Theory
 - The Sticky Price Theory
 - The Misperceptions Theory
 - The three of them have a common theme: Output will deviate from the “natural” level when the price level deviates from the price level that people expected.
 - The difference between observed and expected prices arises because of these market imperfections (sticky wage, sticky price, or misperceptions)

$$\text{Quantity of output supplied} = \text{Natural rate of output} + \alpha^*(\text{Actual price level} - \text{Expected Price level})$$

where α is the responsiveness of Output to changes in the actual price level.

- The Sticky Wage Theory

- Nominal wages are slow to adjust or are “sticky” in the short run:
 - Wages do not adjust immediately to a fall in the price level.
 - A lower price level makes employment and production less profitable.
 - This induces firms to reduce the quantity of goods and services supplied.
 - Because we were expecting higher prices, we fixed higher wages. Now that prices are lower, our costs are too high if we cannot adjust wages, and therefore, we will have to adjust employment and quantities produced

- The Sticky Price Theory

- Prices of some goods and services adjust sluggishly in response to changing economic conditions (due to menu costs, for example).
 - An unexpected price drop leaves some firms with higher-than-desired prices.
 - This depresses sales, which induces firms to reduce the quantity of goods and services they produce.

We were again expecting higher prices, but prices turn out to be lower, and if we cannot adjust prices quickly, we will have less demand, and therefore produce less.

- **The Misperceptions Theory**

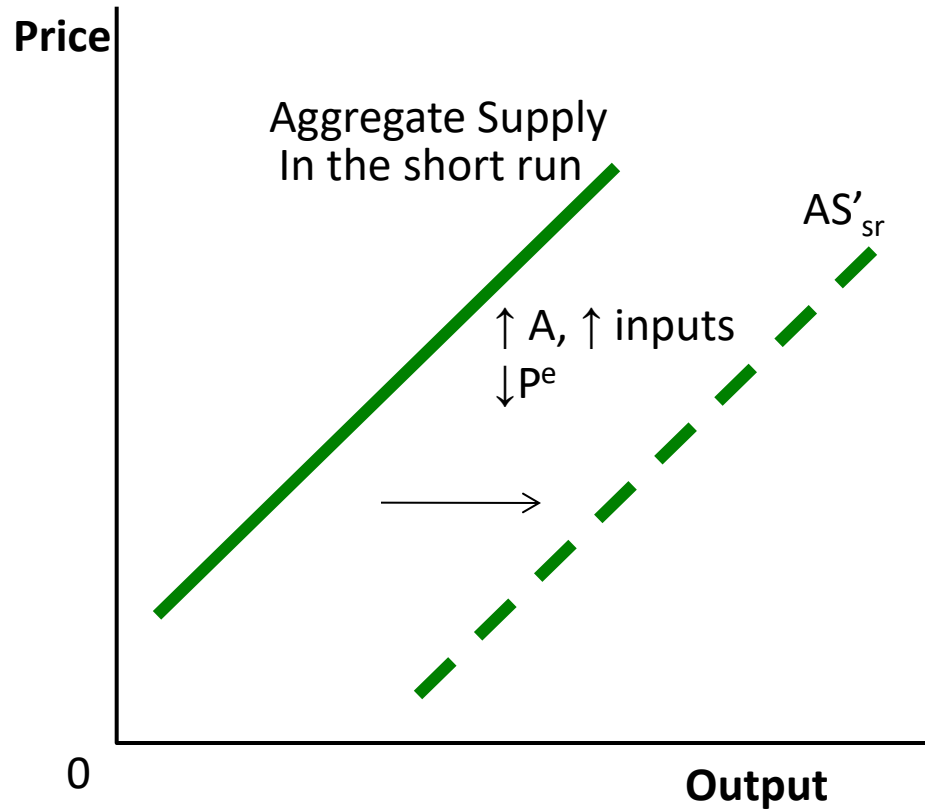
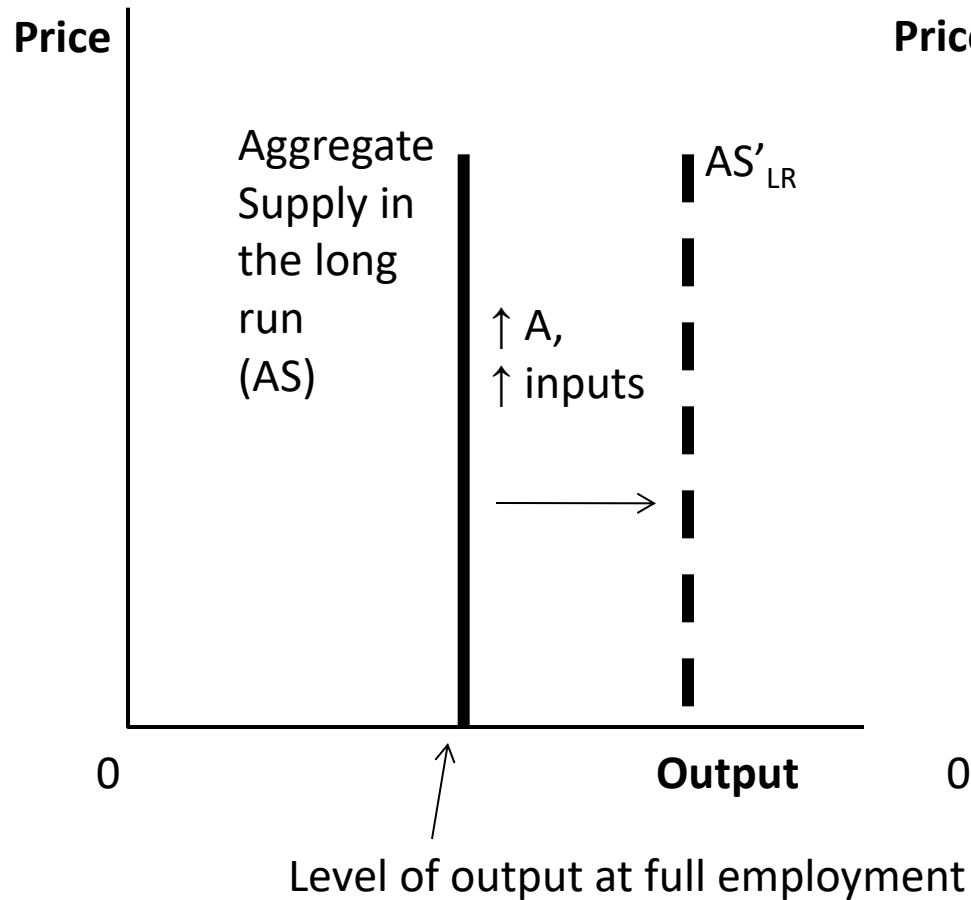
- Changes in the overall price level temporarily mislead suppliers about what is happening in the markets where they sell their output.
- A lower price level causes misperceptions about relative prices.
 - Suppliers may think that it is their price only that is coming down and not all prices
 - These misperceptions induce suppliers to decrease the quantity of goods and services supplied.

We were again expecting higher relative prices, and we now perceive that they are lower, so we will reduce the production of our good

Shifts in the short-run aggregate supply:

- The short-run aggregate supply will shift for the same reasons that the long-run aggregate supply shifts:
$$\text{Short run AS} = \text{Long run AS} + \alpha * (\text{Actual price level} - \text{Expected Price level})$$
- So, changes in labor, physical and human capital, natural resources and technology will lead to shifts in the short-run AS
- But the short run AS will also shift when our expectations about price level change. If we expected price levels to be higher, then this will reduce our AS in the short-run because of the three reasons we saw earlier (wages are too high, we cannot reduce prices in the short run or we think it is only our price going down).
- Changes in expectations are going to play a crucial role in allowing the economy to find its way back to the long-run equilibrium in the long run.
- In the short-run expectations are fixed, but in the long-run expectations will adjust, provoking changes in the short-run AS which will ensure that in the long-run we are back to the long-run equilibrium where long-run aggregate supply equals aggregate demand.

Shifts in AS

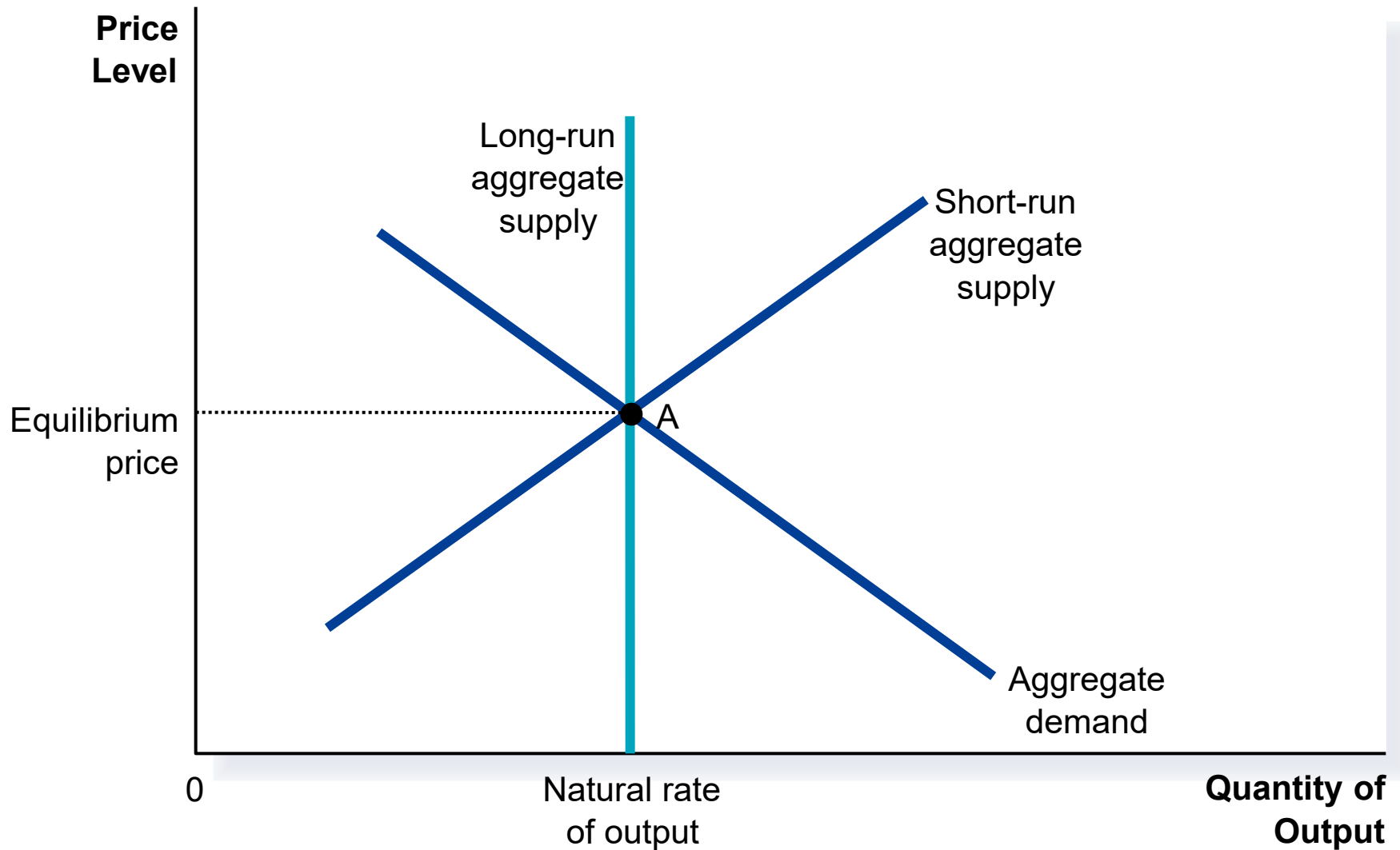


The more flexible prices and wages are,, the more inelastic the AS function.

Chapter Contents

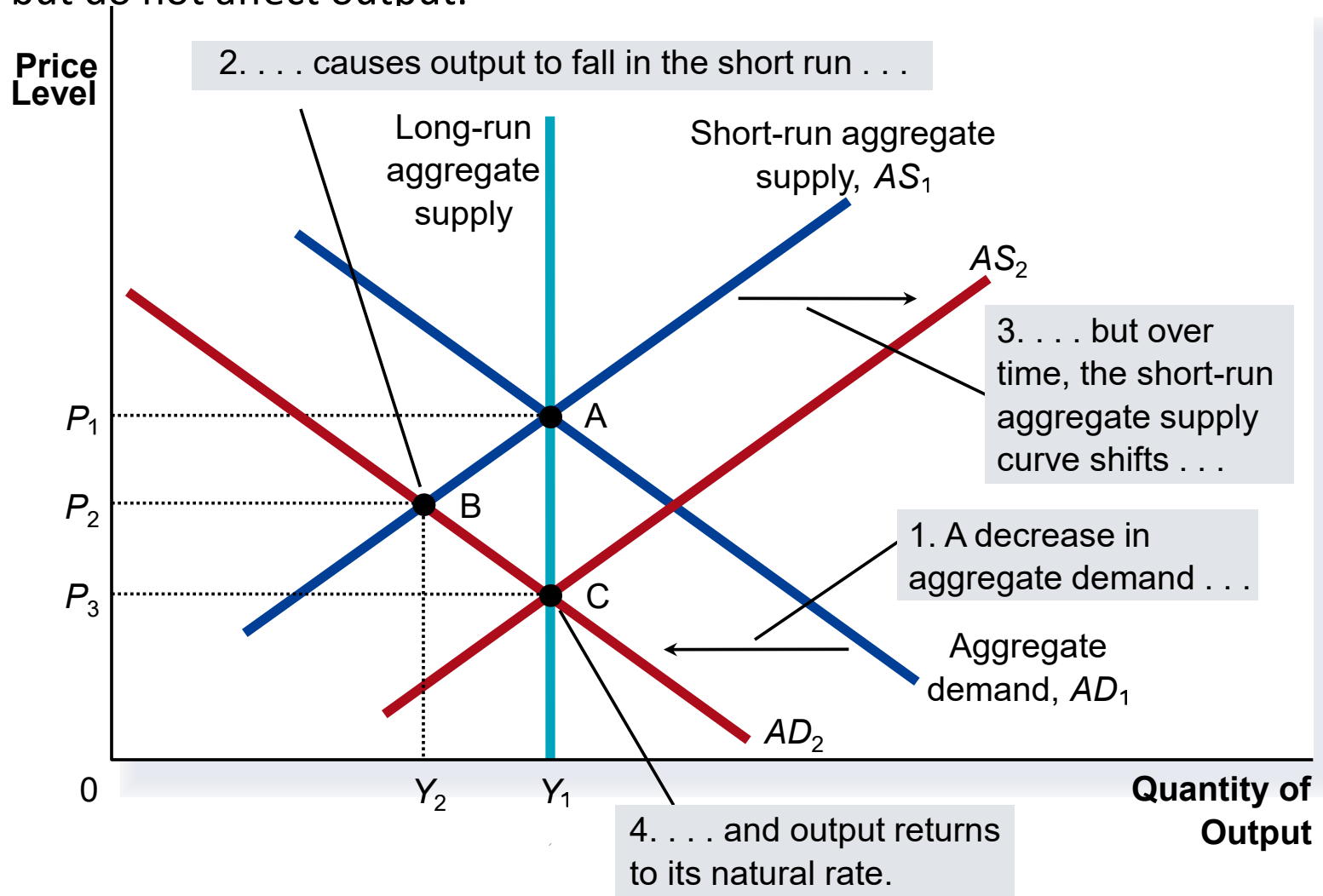
- a) Short-run fluctuations
- b) Aggregate Demand
- c) Aggregate supply
- d) Two main causes of short-run fluctuations
- e) Summary

d. Two main causes of short-run fluctuations



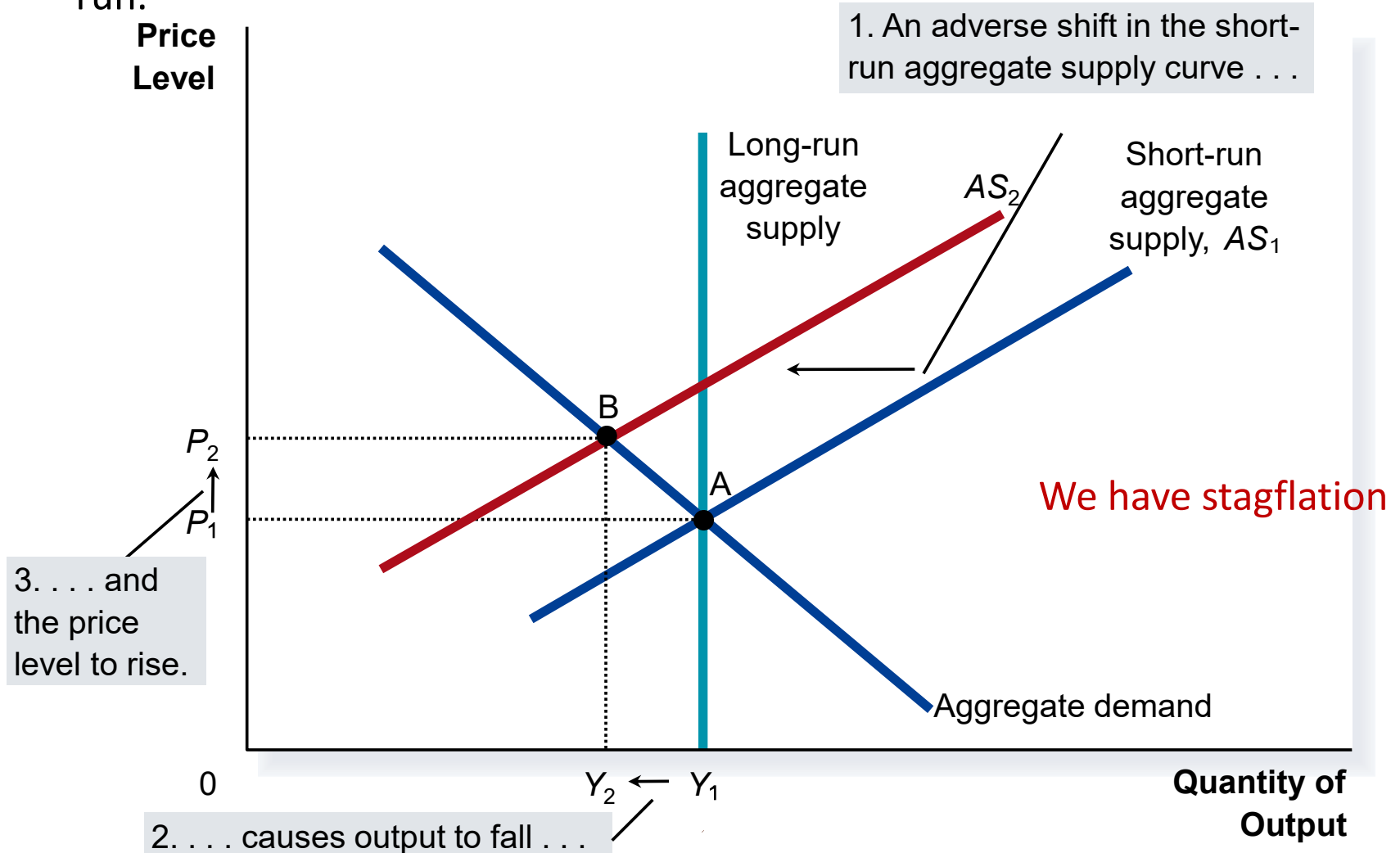
An adverse shift in Aggregate Demand

- In the short run, shifts in aggregate demand cause fluctuations in the economy's output of goods and services.
- In the long run, shifts in aggregate demand affect the overall price level but do not affect output.



An Adverse Shift in Aggregate Supply

- A decrease in one of the determinants of aggregate supply shifts the curve to the left, and therefore output falls below the natural rate of employment, unemployment rises, and the price level rises in the short-run.



Chapter Contents

- a) Short-run fluctuations
- b) Aggregate Demand
- c) Aggregate supply
- d) Two main causes of short-run fluctuations
- e) Summary

e. Summary

- All societies experience short-run economic fluctuations around long-run trends.
- These fluctuations are irregular and largely unpredictable.
- When recessions occur, real GDP and other income measures, spending, and production measures fall, and unemployment rises.
- Economists analyze short-run economic fluctuations using the aggregate demand and aggregate supply model.
- According to the aggregate demand and supply model, the output of goods and services and the overall level of prices adjust to balance aggregate demand and supply.
- The aggregate demand curve slopes downward for three reasons: a wealth effect, an interest rate effect, and an exchange rate effect.
- Any event or policy that changes consumption, investment, government purchases, or net exports at a given price level will shift the aggregate demand curve.

- In the long run, the aggregate supply curve is vertical.
- In the short run, the aggregate supply curve is upward-sloping.
- There are three theories explaining the upward slope of short-run aggregate supply: the misperceptions theory, the sticky wage theory, and the sticky price theory.
- Events that alter the economy's ability to produce output will shift the short-run aggregate supply curve.
- Also, the position of the short-run aggregate supply curve depends on the expected price level.
- One possible cause of economic fluctuations is a shift in aggregate demand.
- A second possible cause of economic fluctuations is a shift in aggregate supply.